

MTH 418a: Inference-I Assignment No. 2: Sufficiency

1. Let $X_1, \dots, X_n$ be a random sample from a population with p.d.f./p.m.f. $f_\theta(\cdot), \theta \in \Theta$. In each of the following cases, using the factorization theorem, show that $T(\underline{X})$ is a sufficient statistic:
 - (a) For known $\mu \in \mathbb{R}$, $X_1 \sim N(\mu, \theta^2)$, $\Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n (X_i - \mu)^2$;
 - (b) $X_1 \sim \text{Beta}(\theta_1, \theta_2)$, $\theta = (\theta_1, \theta_2)$, $\Theta = (0, \infty) \times (0, \infty)$, $T(\underline{X}) = (\prod_{i=1}^n X_i, \prod_{i=1}^n (1 - X_i))$;
 - (c) $X_1 \sim N(\theta, \theta)$, $\Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n X_i^2$;
 - (d) $X_1 \sim N(\theta, \theta^2)$, $\Theta = (0, \infty)$, $T(\underline{X}) = (\bar{X}, \sum_{i=1}^n (X_i - \bar{X})^2)$;
 - (e) $X_1 \sim U(\theta_1, \theta_2)$, $\theta = (\theta_1, \theta_2)$, $\Theta = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 < x_2\}$, $T(\underline{X}) = (X_{(1)}, X_{(n)})$;
 - (f) $X_1 \sim U(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, $\Theta = \mathbb{R}$, $T(\underline{X}) = (X_{(1)}, X_{(n)})$;
 - (g) $f_\theta(x) = \frac{2(\theta-x)}{\theta^2} I_{(0,\theta)}(x)$, $\Theta = (0, \infty)$, $T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$;
 - (h) $X_1 \sim \text{Cauchy}(\theta)$, $\Theta = \mathbb{R}$, $T(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$;
 - (i) $X_1 \sim \text{Poisson}(\theta)$, $\Theta = (0, \infty)$, $T(\underline{X}) = \bar{X}$;
 - (j) For known positive integer m , $X_1 \sim \text{Bin}(m, \theta)$, $\Theta = (0, 1)$, $T(\underline{X}) = \sum_{i=1}^n X_i$;
 - (k) For normalizing constant $c(\theta)$, $f_\theta(x) = \frac{c(\theta)}{x^2}$, $x = \theta+1, \theta+2, \dots$, $\Theta = \{0, 1, 2, \dots\}$, $T(\underline{X}) = X_{(1)}$.
2. Each of the parts (a)-(k) below correspond to the corresponding part of Problem 1. For each of the parts (a)-(k), show that the statistic $U(\underline{X})$, as defined below, is sufficient for θ :
 - (a) $U(\underline{X}) = (\bar{X}, \sum_{i=1}^n (X_i - \bar{X})^2)$;
 - (b) $U(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$;
 - (c) $U(\underline{X}) = 15T^2(\underline{X}) + 8T(\underline{X}) - 23$;
 - (d) $U(\underline{X}) = (\bar{X}, \sum_{i=1}^n (X_i - 1)^2)$;
 - (e) $U(\underline{X}) = (2X_{(n)} - X_{(1)}, X_{(n)} + 3X_{(1)})$;
 - (f) $U(\underline{X}) = (X_{(1)}, X_{(2)}, \dots, X_{(n)})$;
 - (g) $U(\underline{X}) = (\prod_{i=1}^n X_i, X_{(2)} + X_{(3)}, 2X_{(2)} + 3X_{(3)}, X_{(4)}, \dots, X_{(n)})$;
 - (h) $U(\underline{X}) = (X_{(1)}, X_{(2)} - X_{(1)}, X_{(3)} - X_{(2)}, \dots, X_{(n-1)} - X_{(n-2)}, X_{(n)} - X_{(n-1)})$;
 - (i) $U(\underline{X}) = 2\bar{X}^2 + 3\bar{X} - 100$;
 - (j) $U(\underline{X}) = \bar{X}^5$;
 - (k) $U(\underline{X}) = 9e^{2X_{(1)}} - 17e^{X_{(1)}} - 1083$.
3. Let $X_1, \dots, X_n$ be a random sample from a population with p.d.f./p.m.f. $f_\theta(\cdot), \theta \in \Theta$. In each of the following cases, using the conditional distribution of the sample $X_1, \dots, X_n$ given the statistic $T(\underline{X})$, show that the statistic $T(\underline{X})$ is sufficient for θ .
 - (a) $X_1 \sim \text{Poisson}(\theta)$, $\Theta = (0, \infty)$, $T(\underline{X}) = \sum_{i=1}^n X_i$;
 - (b) For known positive integer m , $X_1 \sim \text{Bin}(m, \theta)$, $\Theta = (0, 1)$, $T(\underline{X}) = \sum_{i=1}^n X_i$;
 - (c) $n = 2$, $X_1 \sim U(0, \theta)$, $\Theta = (0, \infty)$, $T(\underline{X}) = \max\{X_1, X_2\}$;
 - (d) $n = 2$, $X_1 \sim N(0, \theta)$, $\Theta = (0, \infty)$, $T(\underline{X}) = X_1^2 + X_2^2$.

4. Let $X_1, \dots, X_n$ be a random sample from a population having p.d.f. $f_\theta(x) = \theta x^{\theta-1}, 0 < x < 1, \theta \in \Theta = (0, \infty)$.
- (a) Show that $T(\underline{X}) = -\sum_{i=1}^n \ln X_i$ is a sufficient statistic for θ ;
- (b) Find the p.d.f. and $T(\underline{X})$;
- (c) Are $T_1(\underline{X}) = (X_1, \dots, X_n)$, $T_2(\underline{X}) = (X_{(1)}, \dots, X_{(n)})$, $T_3(\underline{X}) = (X_1, X_1 + X_2, \prod_{i=3}^n X_i)$, $T_4(\underline{X}) = T^2(\underline{X}) + 3T(\underline{X}) + 2$, $T_5(\underline{X}) = e^{T(\underline{X})} - T(\underline{X}) + \frac{1}{2}$ sufficient statistics for θ ?

MTH 418 A: Inference-I

Assignment No. 2: Sufficiency

Problem No. 1 By the factorization a statistic $T(x)$ is sufficient for $\theta \in \Theta$ iff the joint p.d.f.

$$g_{\theta}(x) = \prod_{i=1}^n b_{\theta}(x_i), \quad x = (x_1, \dots, x_n) \in \mathbb{R}^n, \quad \theta \in \Theta,$$

of $x = (x_1, \dots, x_n)$ can be factorized as

$$g_{\theta}(x) = \psi(\theta, T(x)) h(x), \quad x \in \mathbb{R}^n,$$

for some functions $\psi(\theta, T(x))$ (that depends on θ and $T(x)$) and $h(x)$ (that does not depend on θ).

(a) $T(x) = \sum_{i=1}^n (x_i - \mu)^2$ is sufficient with

$$\psi(\theta, T(x)) = \frac{1}{\theta} e^{-\frac{T(x)}{2\theta}}, \quad h(x) = \left(\frac{1}{2\pi}\right)^{n/2}, \quad x \in \mathbb{R}^n, \quad \theta \in \Theta;$$

(b) $T(x) = \left(\prod_{i=1}^n x_i, \prod_{i=1}^n (1-x_i)\right)$ is sufficient with

$$\psi(\theta, T(x)) = \frac{1}{\theta_1 \theta_2} (T_1(x))^{\theta_1-1} (T_2(x))^{\theta_2-1}, \quad h(x) = \prod_{i=1}^n I_{(0,1)}(x_i),$$

where $T_1(x) = \prod_{i=1}^n x_i$, $T_2(x) = \prod_{i=1}^n (1-x_i)$ and $T(x) = (T_1(x), T_2(x))$

(c) $T(x) = \sum_{i=1}^n x_i^2$ is sufficient with

$$\psi(\theta, T(x)) = \frac{1}{\theta^n} e^{-\frac{n\theta}{2}} e^{-\frac{T(x)}{2\theta}}, \quad h(x) = \left(\frac{1}{2\pi}\right)^{n/2} e^{\sum_{i=1}^n x_i}, \quad x \in \mathbb{R}^n, \quad \theta \in \Theta$$

(d) $T(x) = (\bar{x}, \sum_{i=1}^n (x_i - \bar{x})^2) = (T_1(x), T_2(x))$ is sufficient with

$$\psi(\theta, T(x)) = \left(\frac{1}{2\pi\theta^2}\right)^{n/2} e^{-\frac{1}{2\theta^2}(n(T_1(x)-\theta)^2 + T_2(x))}$$

$$h(x) = \left(\frac{1}{2\pi}\right)^{n/2}, \quad x \in \mathbb{R}^n, \quad \theta \in \Theta$$

(a) $T(\underline{x}) = (x_{(1)}, x_{(n)})$ is sufficient with $\psi(\theta, T(\underline{x})) = \frac{1}{(\theta_2 - \theta_1)^n} I_{(\theta_1, \theta_2)}(x_{(1)}) I_{(\theta_1, \theta_2)}(x_{(n)})$, $h(\underline{x}) = 1$, $\underline{x} \in \mathbb{R}^n$, $\theta \in \Theta$.

(b) $T(\underline{x}) = (x_{(1)}, x_{(n)})$ is sufficient with

$$\psi(\theta, T(\underline{x})) = I_{(\lambda_{(n)} - \frac{1}{2}, \lambda_{(1)} + \frac{1}{2})}(\theta), \quad h(\underline{x}) = 1, \quad \underline{x} \in \mathbb{R}^n, \quad \theta \in \Theta.$$

(g) $T(\underline{x}) = (x_{(1)}, \dots, x_{(n)})$ is sufficient with

$$\psi(\theta, T(\underline{x})) = \left(\frac{2}{\theta_2}\right)^n \prod_{i=1}^n (1 - x_{(i)}) I_{(\lambda_{(n)}, \theta_1)}(\theta), \quad h(\underline{x}) = \prod_{i=1}^n I_{(0, \theta_1)}(x_{(i)})$$

(h) $T(\underline{x}) = (x_{(1)}, \dots, x_{(n)})$ is sufficient with

$$\psi(\theta, T(\underline{x})) = \frac{1}{\prod_{i=1}^n [1 + (x_{(i)} - \theta)^2]}, \quad h(\underline{x}) = \frac{1}{\pi^n}, \quad \underline{x} \in \mathbb{R}^n, \quad \theta \in \Theta.$$

(i) $T(\underline{x}) = \bar{x}$ is sufficient with

$$\psi(\theta, T(\underline{x})) = e^{-n\theta} \theta^{nT(\underline{x})}, \quad h(\underline{x}) = \prod_{i=1}^n \frac{1}{x_i} I_{N_+}(x_i),$$

$\underline{x} \in \mathbb{N}^n, \quad \theta \in \Theta,$

where $N_+ = \{0, 1, 2, \dots\}$

(j) $T(\underline{x}) = \sum_{i=1}^n x_i$ is sufficient with

$$\psi(\theta, T(\underline{x})) = (1-\theta)^n \left(\frac{\theta}{1-\theta}\right)^{T(\underline{x})}, \quad h(\underline{x}) = \prod_{i=1}^n \binom{n}{x_i} I_{\{0, 1, \dots, n\}}(x_i),$$

$\underline{x} \in \mathbb{N}^n, \quad \theta \in \Theta.$

(k) $T(\underline{x}) = x_{(1)}$ is sufficient with

$$\psi(\theta, T(\underline{x})) = (c(\theta))^n I_{(\theta, \lambda_{(1)} - 1)}(\theta), \quad h(\underline{x}) = \prod_{i=1}^n I_{N_+}(x_i).$$

Problem No. 2

We will use the fact that if $T(X)$ is a sufficient statistic and $T(X)$ is a function of statistic $U(X)$, then $U(X)$ is also sufficient.

(a) $T(X) = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2 = n(U_1 - \mu)^2 + U_2$, where $U = (U_1, U_2)$;

(b) $T(X) = \left(\prod_{i=1}^n x_i, \prod_{i=1}^n (1 - x_i) \right) = \left(\prod_{i=1}^n U_i, \prod_{i=1}^n (1 - U_i) \right)$, where $U = (U_1, \dots, U_n)$;

(c) Since $P_\theta(T(X) \geq 0) = 1 \quad \forall \theta \in \Theta$, $U(X) = 15T^2(X) + 8T(X) - 23$ is a strictly increasing function of $T(X)$ and $T(X) = U^1(X)$;

(d) $T(X) = \left(\bar{x}, \sum_{i=1}^n (x_i - \bar{x})^2 \right)$
 $= \left(\bar{x}, \sum_{i=1}^n (x_i - 1)^2 - n(\bar{x} - 1)^2 \right) = (U_1, U_2 - n(U_1 - 1)^2)$, where $U = (U_1, U_2)$;

(e) $T(X) = (x_{(1)}, x_{(n)}) = \left(\frac{2U_2 - U_1}{7}, \frac{3U_1 + U_2}{7} \right)$, where $U = (U_1, U_2)$;

(f) $T(X) = (U_1, U_n)$, where $U = (U_1, \dots, U_n)$;

(g) $T(X) = \left(\frac{U_1}{3U_2 - U_3}, 3U_2 - U_3, U_3 - 2U_2, U_4, \dots, U_n \right)$, where $U = (U_1, \dots, U_n)$;

(h) $T(X) = (U_1, U_1 + U_2, U_1 + U_2 + U_3, \dots, U_1 + U_2 + \dots + U_n)$, where $U = (U_1, \dots, U_n)$;

(i) Since $P_\theta(\bar{X} \geq 0) = 1 \quad \forall \theta \in \Theta$, $T(X) = (U(X))^{75}$
 Since $P_\theta(\bar{X} \geq 0) = 1 \quad \forall \theta \in \Theta$, $U(X) = 2\bar{X}^2 + 3\bar{X} - 100$ is a strictly increasing function of $\bar{X}$ and $\bar{X} = U^1(X)$.

(j) Since $P_\theta(X_{(1)} \geq 1) = 1 \quad \forall \theta \in \Theta$ and

$\frac{dU}{dX_{(1)}} = 18e^{2X_{(1)}} - 17 \geq 0 \quad \forall \theta \in \Theta$,

$U(X)$ is a strictly increasing function of $\bar{X}$ and $T(X) = X_{(1)} = U^1(X)$

Problem 410.3

(a) For $t \in \{0, 1, 2, \dots\}$

$$P_0(x_1 = \lambda_1, x_2 = \lambda_2, \dots, x_n = \lambda_n | T = t)$$

$$= \frac{P_0(x_1 = \lambda_1, x_2 = \lambda_2, \dots, x_n = \lambda_n, \sum_{i=1}^n x_i = t)}{P_0(\sum_{i=1}^n x_i = t)}$$

$$= \begin{cases} \frac{P_0(x_1 = \lambda_1, \dots, x_n = \lambda_n)}{P_0(\sum_{i=1}^n x_i = t)}, & \text{if } \sum_{i=1}^n \lambda_i = t \\ 0, & \text{otherwise.} \end{cases}$$

$$= \begin{cases} \frac{\prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}}{\frac{e^{-n\theta} (n\theta)^t}{L^t}}, & \lambda_i = 0, 1, 2, \dots, i=1, \dots, n \text{ and } \sum_{i=1}^n \lambda_i = t \\ 0, & \text{otherwise} \end{cases}$$

($\sum_{i=1}^n x_i \sim \text{Poisson}(n\theta)$)

$$= \begin{cases} \frac{L^t}{(\prod_{i=1}^n x_i!) n^t}, & \lambda_i = 0, 1, 2, \dots, i=1, \dots, n \text{ and } \sum_{i=1}^n \lambda_i = t \\ 0, & \text{otherwise} \end{cases}$$

→ does not depend on $\theta \in \mathbb{R}^+$.

(b) For $t \in \{0, 1, \dots, mn\}$

$$P_0(X_1 = \lambda_1, \dots, X_n = \lambda_n \mid T = t)$$

$$= \frac{P_0(X_1 = \lambda_1, \dots, X_n = \lambda_n, \sum_{i=1}^n X_i = t)}{P_0(\sum_{i=1}^n X_i = t)}$$

$$= \begin{cases} \frac{P_0(X_1 = \lambda_1, \dots, X_n = \lambda_n)}{P_0(\sum_{i=1}^n X_i = t)} \\ 0 \end{cases}$$

, $\lambda_i = 0, 1, \dots, m$ ($i = 1, \dots, n$),
and $\sum_{i=1}^n \lambda_i = t$

otherwise

$$= \begin{cases} \frac{\prod_{i=1}^n \binom{m}{\lambda_i} \theta^{\lambda_i} (1-\theta)^{m-\lambda_i}}{\binom{mn}{t} \theta^t (1-\theta)^{mn-t}}, & \lambda_i = 0, 1, \dots, m \text{ ($i = 1, \dots, n$)}, \\ & \text{and } \sum_{i=1}^n \lambda_i = t \\ 0 \end{cases}$$

otherwise

$$= \begin{cases} \frac{\prod_{i=1}^n \binom{m}{\lambda_i}}{\binom{mn}{t}}, & \lambda_i = 0, 1, \dots, m \text{ ($i = 1, \dots, n$)}, \\ & \text{and } \sum_{i=1}^n \lambda_i = t \\ 0 \end{cases}$$

otherwise

→ does not depend on $\theta \in (0, 1)$

(c) For $t \in (0, \infty)$

$$F_{X_1, X_2 | T}(x_1, x_2 | t) = P_0(X_1 \leq x_1, X_2 \leq x_2 | T = t)$$

$$= \lim_{h \downarrow 0} P_0(X_1 \leq x_1, X_2 \leq x_2 | t-h < T \leq t)$$

$$= \lim_{h \downarrow 0} \frac{P_0(X_1 \leq x_1, X_2 \leq x_2, t-h < T \leq t)}{P_0(t-h < T \leq t)}$$

$$= \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{P_0(X_1 \leq x_1, X_2 \leq x_2, t-h < T \leq t)}{h}$$

$$= \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{P_0(X_1 \leq x_1, X_2 \leq x_2, T \leq t) - P_0(X_1 \leq x_1, X_2 \leq x_2, T \leq t-h)}{h}$$

$$= \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{F(\min\{x_1, t\}) F(\min\{x_2, t\}) - F(\min\{x_1, t-h\}) F(\min\{x_2, t-h\})}{h}$$

$$= \begin{cases} \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{F(x_1)F(x_2) - F(x_1)F(x_2)}{h}, & x_1 < t, x_2 < t \\ \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{F(x_1)F(t) - F(x_1)F(t-h)}{h}, & x_1 < t, x_2 \geq t \\ \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{F(t)F(x_2) - F(t-h)F(x_2)}{h}, & x_1 \geq t, x_2 < t \\ \frac{1}{b_T(t)} \lim_{h \downarrow 0} \frac{F(t)F(t) - F(t-h)F(t-h)}{h}, & x_1 \geq t, x_2 \geq t \end{cases}$$

$$= \begin{cases} 0, & x_1 < t, x_2 < t \\ \frac{F(x_1)b(t)}{b_T(t)}, & x_1 < t, x_2 \geq t \\ \frac{F(x_2)b(t)}{b_T(t)}, & x_1 \geq t, x_2 < t \\ \frac{2F(t)b(t)}{b_T(t)}, & x_1 \geq t, x_2 \geq t \end{cases}$$

$$= \begin{cases} 0, & \lambda_1 < t, \lambda_2 < t \\ \frac{\lambda_1}{2t}, & \lambda_1 < t, \lambda_2 \geq t \\ \frac{\lambda_2}{2t}, & \lambda_1 \geq t, \lambda_2 < t \\ 1, & \lambda_1 \geq t, \lambda_2 \geq t \end{cases}$$

→ does not depend on θ .

$$b_{T|t+1} = 2(F(t+1))b_{t+1} \\ = \begin{cases} \frac{2t}{\theta^2}, & 0 < t < \theta \\ 0, & \text{o.w.} \end{cases}$$

Note: The density of $(x_1, x_2) | T=t$ is given by $0 < t < \theta$

$$P(x_1 = t, x_2 \leq \lambda_2 | T=t) = P(x_1 \leq t, x_2 \leq \lambda_2 | T=t) - P(x_1 < t, x_2 \leq \lambda_2 | T=t) \\ = \begin{cases} \frac{\lambda_2}{2t}, & 0 < \lambda_2 < t \\ 1, & \lambda_2 \geq t \end{cases}$$

$$P(x_1 \leq \lambda_1, x_2 = t | T=t) = P(x_1 \leq \lambda_1, x_2 \leq t | T=t) - P(x_1 \leq \lambda_1, x_2 < t | T=t) \\ = \begin{cases} \frac{\lambda_1}{2t}, & 0 < \lambda_1 < t \\ 1, & \lambda_1 \geq t \end{cases}$$

$$b_{x_1, x_2 | T}(\lambda_1, \lambda_2 | t) = \begin{cases} \frac{1}{2t}, & \lambda_1 = t, 0 < \lambda_2 < t \\ & \text{or} \\ & 0 < \lambda_1 < t, \lambda_2 = t \\ 0, & \text{otherwise} \end{cases}$$

(d) The joint p.d.f. of $\underline{x} = (x_1, x_2)$ is
 $g(\lambda_1, \lambda_2) = \frac{1}{2\pi\theta} e^{-\frac{1}{2\theta}(\lambda_1^2 + \lambda_2^2)}, \quad -\theta < \lambda_i < \theta \quad (i=1,2)$

Consider r.v.s defined through polar coordinates of (x_1, x_2) as

$$x_1 = R \cos \Theta, \quad x_2 = R \sin \Theta$$

$$x_1 = r \cos \theta, \quad x_2 = r \sin \theta, \quad r > 0, \quad 0 < \theta < 2\pi$$

$$J = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

7/9

The joint p.d.f of (R, Θ) is

$$f_{R, \Theta}(r, \theta) = \frac{1}{2\pi\theta} \times r e^{-\frac{r^2}{2\theta}}, \quad r > 0, \quad 0 < \theta < 2\pi$$

$$\Rightarrow R \sim U(0, 2\pi) \text{ and } \Theta \mid (f_{\Theta|\theta}(\theta) = \frac{r e^{-\frac{r^2}{2\theta}}}{\theta}, \quad r > 0)$$

are independently distributed

For $t > 0$

$$(x_1, x_2) \mid x_1^2 + x_2^2 = t \stackrel{d}{=} (R \cos \Theta, R \sin \Theta) \mid R^2 = t$$

$$\stackrel{d}{=} (\sqrt{t} \cos \Theta, \sqrt{t} \sin \Theta) \mid R = \sqrt{t}$$

$$\stackrel{d}{=} (\sqrt{t} \cos \Theta, \sqrt{t} \sin \Theta) \quad (\text{as } R \text{ and } \Theta \text{ are independently distributed})$$

Since the distribution of $\Theta \mid (\sim U(0, 2\pi))$ does not depend on θ , the distribution of $(x_1, x_2) \mid x_1^2 + x_2^2 = t$ does not depend on θ .

Problem 11.4

(a) The joint p.d.f. of $\underline{x} = (x_1, \dots, x_n)$ is

$$g_{\theta}(x_1, \dots, x_n) = \theta^n \left(\prod_{i=1}^n x_i \right)^{\theta-1} \prod_{i=1}^n I_{(0,1)}(x_i)$$

$$= \theta^n e^{(1-\theta)T(\underline{x})} \prod_{i=1}^n I_{(0,1)}(x_i)$$

$$= \psi(\theta, T(\underline{x})) h(\underline{x})$$

$$\text{where } \psi(\theta, T(\underline{x})) = \theta^n e^{-\theta T(\underline{x})}, \quad h(\underline{x}) = e^{T(\underline{x})} \prod_{i=1}^n I_{(0,1)}(x_i)$$

By the factorization theorem $T(\underline{x}) = -\sum_{i=1}^n \ln x_i$ is sufficient for θ .

(b) Let $y_i = -\ln x_i$, $i=1, \dots, n$. Then $y_1, y_2, \dots, y_n$ are iid with p.d.f.

$$f_{y_i}(y) = \theta e^{-\theta y}, \quad y > 0, \quad i=1, \dots, n$$

$$\rightarrow \text{Gamma}(1, \frac{1}{\theta})$$

$$\Rightarrow T(\underline{x}) = \sum_{i=1}^n T_i \sim \text{Gamma}(n, \frac{1}{\theta}).$$

The p.d.f. of T is

$$f_{T|H_1} = \frac{\theta^n}{\Gamma(n)} e^{-\theta \cdot T} T^{n-1} > 0.$$

(c) • Since the sufficient statistic $T(\underline{x}) = -\sum_{i=1}^n \ln x_i$ is a function of $T_1(\underline{x}) = (x_1, \dots, x_n)$, the statistic $T_1(\underline{x})$ is also sufficient;

• Since the sufficient statistic $T(\underline{x}) = -\sum_{i=1}^n \ln x_i = -\sum_{i=1}^n \ln(x_{(i)})$ is a function of $T_2(\underline{x}) = (x_{(1)}, \dots, x_{(n)})$, the statistic $T_2(\underline{x})$ is also sufficient;

$$T(\underline{x}) = -\ln\left(\prod_{i=1}^n x_i\right) = -\ln(u_1(u_2 - u_1) \prod_{i=3}^n u_i),$$

where $T_3(\underline{x}) = (u_1, u_2, \dots, u_n)$

Thus sufficient statistic $T(\underline{x})$ is a function of $T_3(\underline{x})$ and consequently $T_3(\underline{x})$ is also sufficient.

• Since $P_\theta(T(\underline{x}) \geq 0) = 1 \quad \forall \theta \in \Theta$, and

$$\frac{dT_4(\underline{x})}{dT(\underline{x})} = 2T(\underline{x}) + 3 > 0,$$

$T_4(\underline{x})$ is a strictly increasing function of $T(\underline{x})$.
(T and T_4 are in 1-1 correspondence)
 $T(\underline{x})$ and $T_4(\underline{x}) = T_4(\underline{x})$

$\Rightarrow T_4(\underline{x})$ is sufficient

• Since $P_\theta(T(\underline{x}) \geq 0) = 1, \quad \forall \theta \in \Theta$ and

$$\frac{dT_5(\underline{x})}{dT(\underline{x})} = e^{T(\underline{x})} - 1 > 0,$$

it follows that T_5 and T are in 1-1 correspondence

$\Rightarrow T_5(\underline{x})$ is sufficient.